AQA GCSE BIOLOGY

Chapter 5: Homeostasis and Response

COMPLETE REVISION NOTES | Higher Tier

Topics Covered

Homeostasis | Nervous System | Synapses | Hormonal Coordination
Menstrual Cycle | Contraception | Blood Glucose Regulation
Diabetes | Thermoregulation | Kidney & Osmoregulation

1. Homeostasis

Homeostasis is the maintenance of a stable internal environment despite changes in external conditions. It is essential for cells to function optimally — enzymes require specific temperature and pH, and cells need the right water potential.

What is Controlled?

Variable Controlled	Why It Matters
Body temperature (37°C)	Enzymes work at optimum rate; prevents denaturation
Blood glucose concentration	Provides cells with energy; prevents osmotic damage
Water content (osmoregulation)	Maintains cell turgor; prevents lysis or crenation
Ion content (Na+, K+)	Required for nerve impulses and muscle contractions
Carbon dioxide levels	Prevents pH changes in the blood

How Homeostasis Works — The Control System

All homeostatic systems use a negative feedback mechanism:

- Receptor — detects a change (stimulus) in the internal environment
- Coordination centre — receives information and processes it (e.g. brain, pancreas)
- Effector — produces a response to restore normal levels (e.g. muscle, gland)
- Negative feedback — the response reverses the change back to the set point

⚠EXAMINER WARNING

Students frequently confuse positive and negative feedback. In AQA Biology, homeostasis always uses NEGATIVE feedback — the response opposes the change. Never say it 'increases the change'.

2. The Nervous System

The nervous system allows rapid, short-lived responses to stimuli. It uses electrical impulses carried along neurones.

Structure of the Nervous System

Division	Components & Function
Central Nervous System (CNS)	Brain + spinal cord. Processes information and coordinates responses
Peripheral Nervous System (PNS)	All nerves outside the CNS. Carries signals to and from the CNS
Sensory neurones	Carry impulses FROM receptors TO CNS
Motor neurones	Carry impulses FROM CNS TO effectors (muscles/glands)
Relay neurones	Connect sensory to motor neurones within the CNS

Types of Receptors

Receptor	Stimulus Detected
Photoreceptors (retina)	Light
Chemoreceptors	Chemicals (e.g. taste, smell, blood CO2)
Mechanoreceptors	Pressure / touch / sound
Thermoreceptors	Temperature changes
Pain receptors (nociceptors)	Tissue damage

The Reflex Arc

A reflex arc is a rapid, automatic, involuntary response that bypasses the brain. This protects the body from harm.

- Stimulus detected by receptor
- Impulse travels along sensory neurone to spinal cord
- Relay neurone in spinal cord connects to motor neurone
- Impulse travels along motor neurone to effector
- Effector (muscle or gland) produces response

Reflex Arc Diagram

Stimulus(e.g. heat)	→	**Receptor(skin)**	→	**SensoryNeurone**	→	**RelayNeurone(spinal cord)**	→	**Motor Neurone+ Effector(response)**

⚠EXAMINER WARNING
Exam tip: Reflex actions do NOT involve the brain — impulses go to the SPINAL CORD. The brain becomes aware of the reflex AFTER it happens. Never write that the brain causes a reflex.

3. Synapses

Structure and Function

A synapse is the junction between two neurones. Electrical impulses cannot cross the gap — instead, chemical signals called neurotransmitters are used.

- Nerve impulse reaches presynaptic neurone terminal
- Vesicles containing neurotransmitter fuse with presynaptic membrane
- Neurotransmitter is released into the synaptic cleft by exocytosis
- Neurotransmitter diffuses across the cleft
- Binds to complementary receptors on the postsynaptic membrane
- New electrical impulse is triggered in the postsynaptic neurone
- Neurotransmitter is broken down by enzymes or reabsorbed

⚠EXAMINER WARNING
Students often say impulses 'travel' across a synapse. They do NOT — neurotransmitters DIFFUSE across. The impulse stops and is recreated on the other side.

Drugs and Synapses

Drug Type	Effect at Synapse
Stimulants (e.g. caffeine, nicotine)	Increase neurotransmitter release or mimic neurotransmitter — MORE impulses
Depressants (e.g. alcohol)	Block receptors or reduce neurotransmitter release — FEWER impulses
Painkillers (e.g. morphine)	Block pain receptors — prevent pain signals reaching brain
Ecstasy (MDMA)	Prevents reuptake of serotonin — builds up in synapse, enhances mood

4. Hormonal Coordination

The endocrine system uses hormones — chemical messengers secreted by glands — carried in the blood to target organs. It is slower and longer-lasting than the nervous system.

Key Endocrine Glands and Hormones

Gland	Hormone(s)	Function
Pituitary gland	FSH, LH, ADH, growth hormone	Master gland — controls other glands
Thyroid gland	Thyroxine	Controls metabolic rate; important in development
Adrenal glands	Adrenaline	Prepares body for 'fight or flight' response
Pancreas	Insulin, Glucagon	Regulates blood glucose concentration
Ovaries	Oestrogen, Progesterone	Controls female reproductive cycle
Testes	Testosterone	Controls male secondary sexual characteristics

Nervous System vs Endocrine System

Nervous System	Endocrine System
Fast response (milliseconds)	Slow response (seconds to days)
Short-lived effect	Long-lasting effect
Electrical impulse	Chemical hormone in blood
Specific target (neurone/effector)	Widespread effect via bloodstream
Does not travel in blood	Transported in blood to target organ

5. The Menstrual Cycle

The menstrual cycle is a monthly cycle of hormonal changes that prepares the uterus for pregnancy. It is controlled by four key hormones.

Hormone	Where Produced	Main Function
FSH (Follicle Stimulating Hormone)	Pituitary gland	Stimulates follicle maturation; stimulates oestrogen production

Oestrogen	Ovaries (follicle)	Repairs uterus lining; inhibits FSH; stimulates LH surge
LH (Luteinising Hormone)	Pituitary gland	Triggers ovulation on day ~14
Progesterone	Corpus luteum (ovary)	Maintains uterus lining; inhibits FSH and LH

Cycle Summary

Days	Key Events
Day 1-5	Menstruation — uterus lining breaks down
Day 1-13	FSH rises; follicle matures; oestrogen rises; lining rebuilds
Day 13-14	LH surge triggers ovulation — egg released from follicle
Day 15-28	Corpus luteum forms; progesterone rises; lining maintained
Day 28	If no fertilisation: corpus luteum breaks down; progesterone falls; menstruation begins again

⚠EXAMINER WARNING

'Inhibits FSH' is an example of negative feedback. Oestrogen rises, which inhibits further FSH production — preventing more follicles from developing. This is a highly examined negative feedback loop.

6. Contraception

Hormonal Contraception

Method	How It Works
Combined pill (oestrogen + progesterone)	Inhibits FSH — prevents egg maturation. Also thickens cervical mucus.
Progesterone-only pill (mini pill)	Thickens cervical mucus; may inhibit ovulation
Hormonal implant / injection	Releases progesterone to prevent ovulation for months/years
Hormonal IUD (Mirena coil)	Releases progesterone locally — thickens mucus, thins lining

Non-Hormonal Contraception

Method	How It Works
Barrier methods (condom, diaphragm)	Physically prevent sperm reaching egg
Copper IUD (non-hormonal)	Copper ions toxic to sperm; also prevents implantation
Spermicide	Chemical that kills or immobilises sperm
Abstinence / natural family planning	Avoiding sex during fertile period — least reliable method
Sterilisation (vasectomy/tubal ligation)	Permanent surgical cutting of vas deferens or fallopian tubes

7. Blood Glucose Regulation

Blood glucose concentration must stay within a narrow range. The pancreas monitors glucose levels and releases hormones to correct any deviation.

Role of the Pancreas

Condition	Response
Blood glucose TOO HIGH (after a meal)	Beta cells in islets of Langerhans secrete INSULIN
What insulin does	Causes cells to absorb glucose from blood; excess glucose converted to GLYCOGEN in liver (glycogenesis)
Blood glucose TOO LOW (during exercise/fasting)	Alpha cells in islets of Langerhans secrete GLUCAGON
What glucagon does	Stimulates liver to convert GLYCOGEN back to GLUCOSE (glycogenolysis) — released into blood

⚠EXAMINER WARNING

Glucagon does NOT produce glucose from scratch — it converts STORED glycogen back to glucose. The word is glycogenolysis. Many students write 'glucagon makes glucose' — this is imprecise and will lose marks.

Diabetes

Type 1 Diabetes	Type 2 Diabetes
Autoimmune: immune system destroys beta cells	Body cells become resistant to insulin

Little or no insulin produced	Pancreas may still produce insulin but cells don't respond
Genetic + autoimmune cause	Linked to diet, obesity, lifestyle (partly genetic)
Managed with insulin injections/pump	Managed with diet, exercise, medication (e.g. metformin)
Cannot be prevented	Can often be prevented by healthy lifestyle

8. Thermoregulation

Core body temperature is maintained at 37°C by the thermoregulatory centre in the hypothalamus.

Temperature Receptors

- Peripheral thermoreceptors in the skin detect external temperature
- Central thermoreceptors in the hypothalamus detect blood temperature
- Hypothalamus acts as coordination centre and effector controller

Responses to OVERHEATING

Mechanism	How It Cools the Body
Sweating	Sweat evaporates from skin surface — takes heat energy away from body (latent heat of evaporation)
Vasodilation	Arterioles near skin surface widen — more blood flows to skin — heat lost by radiation
Erector muscles relax	Body hair lies flat — less air trapped — less insulation (less effective in humans)

Responses to GETTING TOO COLD

Mechanism	How It Warms the Body
Shivering	Rapid muscle contractions generate heat through respiration
Vasoconstriction	Arterioles near skin surface narrow — less blood to skin — less heat lost
Erector muscles contract	Hair stands on end — traps air layer — insulation (more effective in other mammals)
Increased metabolic rate	More respiration produces more heat (controlled by thyroxine)

⚠EXAMINER WARNING

'Vasodilation means more blood goes to the skin' — always explain the consequence: MORE heat lost by radiation. Just saying 'blood vessels widen' is only 1 mark out of 2.

9. The Kidneys and Osmoregulation

The kidneys filter the blood and regulate water content (osmoregulation). They produce urine as a waste product.

Functions of the Kidney

- Filtration of blood — removes urea, excess water, excess ions
- Reabsorption — returns useful substances back to blood (glucose, water, some ions)
- Excretion — removes urea (from protein breakdown) in urine
- Osmoregulation — regulates water content of blood

ADH and Water Regulation

ADH (Anti-Diuretic Hormone) is secreted by the pituitary gland and controls how much water is reabsorbed by the kidneys.

Condition	ADH Response
Blood water content TOO LOW (dehydrated)	MORE ADH released — kidney tubules MORE permeable — MORE water reabsorbed — SMALL volume of concentrated urine
Blood water content TOO HIGH (overhydrated)	LESS ADH released — kidney tubules LESS permeable — LESS water reabsorbed — LARGE volume of dilute urine

⚠EXAMINER WARNING

ADH is always a negative feedback loop. Dehydration → more ADH → more water reabsorbed → blood water increases → less ADH. Students lose marks by not completing the feedback loop explanation.

Kidney Failure Treatment

Treatment	How It Works / Key Points
Dialysis (haemodialysis)	Blood filtered through dialysis machine. Fluid has same concentration as normal blood — only waste diffuses out. Regular sessions required (3x/week, 4hr+)

Kidney transplant	Donor kidney implanted. Immunosuppressant drugs prevent rejection. Long-term solution but risk of rejection and infection
Peritoneal dialysis	Fluid introduced into abdominal cavity. Waste diffuses across peritoneum lining. Done at home but infection risk

10. Key Terms & Formulae

Term	Definition	Examiner Note
Homeostasis	Maintaining stable internal conditions	Must say 'despite external changes'
Negative feedback	Response that reverses a change back to set point	Always reverses — NOT enhances
Neurotransmitter	Chemical that transmits impulse across a synapse	DIFFUSES — not carried electrically
Hormone	Chemical messenger carried in blood to target organ	Slower and longer-lasting than nerve impulse
Insulin	Hormone that lowers blood glucose by promoting glycogenesis	Made by BETA cells in islets of Langerhans
Glucagon	Hormone that raises blood glucose by promoting glycogenolysis	Made by ALPHA cells — converts GLYCOGEN to glucose
ADH	Hormone that increases kidney tubule permeability to water	More ADH = more concentrated urine
Vasodilation	Widening of arterioles near skin — increases heat loss	Must state consequence: MORE heat lost by radiation
Glycogenesis	Conversion of glucose to glycogen (storage)	Insulin stimulates this process
Glycogenolysis	Conversion of glycogen back to glucose	Glucagon stimulates this process

End of Revision Notes — Proceed to Worksheets